

In this Standard the comma has been adopted as decimal sign.

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1. SCOPE

This Standard establishes the requirements for manufacturing, inspection and qualification of parts in composite materials, manufactured by Hand Lay-up or Automated processes (ATL or AFP), as specified on the applicable drawings.

2. APPLICATION

This Standard applies to all parts of composite materials manufactured at Embraer or by Sub-Contractors.

3. APPLICABLE DOCUMENTS

Embraer Standards

MEP 02-009	- Lightning Wire Cloth and Expanded Foil.
MEP 06-029	- Polyvinyl Fluoride Film ("Tedlar®").
MEP 09-063	- Structural Adhesive Film - 121°C (250°F) and 177°C (350°F) Curing.
MEP 10-051	- Compound, Honeycomb Reinforcement.
MEP 15-006	- Glass Fabrics Preimpregnated with Epoxy Resin 121°C (250°F) - Curing.
MEP 15-007	- Aramid Fabrics Preimpregnated with Epoxy Resin 121°C (250°F) - Curing.
MEP 15-008	- Glass Fabrics Preimpregnated with Epoxy Resin 177°C (350°F) - Curing.
MEP 15-009	- Aramid Fabrics and Tape Preimpregnated with Epoxy Resin 177°C (350°F) - Curing.
MEP 15-010	- Nonmetallic Honeycomb Core Material.
MEP 15-014	- Carbon Fabrics and Tape Preimpregnated with Epoxy Resin, 177°C (350°F) - Curing.
MEP 15-016	- Glass Fabrics Preimpregnated with Phenolic Resin, 177°C (350°F) - Curing.
MEP 15-018	- Carbon Fabrics and Tape Pre-impregnated with Epoxy Resin 121°C (250°F) - Curing.
MEP 15-022	- Carbon Fabrics and Tape Pre-impregnated with Impact-Resistant Epoxy Resin 177°C (350°F) - Curing.
MEP 15-025	- Carbon Fabrics Preimpregnated with Polyimide Resin, 177°C (350°F) - Curing.
MEP 15-026	- Glass Fabrics Preimpregnated with Phenolic Resin 121°C (250°F) - Curing.
MEP 15-027	- Aramid Fabrics Pre-impregnated with Phenolic Resin, 121°C (250°F) - Curing.
MEP 15-028	- Carbon Fabrics and Tape Pre-impregnated with Phenolic Resin, 121°C (250°F) - Curing.
MEP 15-047	- Carbon Fabric and Tape Preimpregnated With Epoxy Resin, 180°C (355°F) - Curing
MEP 15-054	- Carbon Tape Preimpregnated with Epoxy Resin, 180°C (355°F) Curing.
MEP 15-055	- Epoxy Resin Film 177°C (350°F) Curing for Resin Film Infusion (RFI) Process.
MEP 15-057	- Carbon Fiber Reinforcement For Liquid Composite Molding.

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MEP 15-059	- Woven Intermediate Modulus Carbon Preimpregnated with Epoxy Resin 180°C (355°F) - Curing.
MEP 15-060	- Glass Fabrics Pre-impregnated with Epoxy Resin 180°C (355°F) Curing.
MEP 15-063	- Intermediate Modulus Carbon Fiber NCF (Non Crimp Fabric) Applicable in the Resin Film Infusion (RFI) Process.
MEP 15-064	- Carbon Fiber Pre-impregnated with Epoxy Resin, Out of Autoclave Curing.
MEP 15-067	- Carbon Tape Preimpregnated with Toughened Epoxy Resin, 177°C (350°F) Curing.
MEP 15-068	- Glass Fabrics Preimpregnated with Toughened Epoxy Resin, 177°C (350°F) Curing.
MEP 15-070	- Woven Intermediate Modulus Carbon Preimpregnated with Toughened Epoxy Resin, 177°C (350°F) Curing.
MEP 15-071	- Carbon Fabric Pre-impregnated with Epoxy Resin, 121°C (250°F) - Curing.
MEP 22-013	- Epoxy Resin for Liquid Infusion.
NE 01-103	- Parts - Marking Methods.
NE 01-107	- Serialized Parts, GSE and Aircraft - Marking of.
NE 01-114	- Serialization of EMB-145 Manufactured Parts and Aircraft Marking.
NE 01-121	- Serialization of ERJ 170/190 Manufactured Parts and Aircraft Marking.
NE 03-047	- Elastomer, Plastic, Composite Material, Interior Monuments Parts - General Tolerance.
NE 06-042	- Sandwich Panels - Insert Installation, Filling of Voids and Protection of Exposed Core (Edges Protection).
NE 07-002	- Limited Contamination Areas - Environment Requirements and Controls.
NE 07-009	- Composite Materials - Visual Inspection.
NE 07-011	- Nondestructive Inspection of Composite and Bonded Parts.
NE 20-038	- Repairs of Composite Material Parts.
NE 40-077	- Equipment for Thermal Processing - Temperature Control and Classification.
NE 40-188	- Composite Laminate, Curing at 177°C (350°F) / 30 psi - Autoclave Process - Requirements and Manufacturing Procedures.
NE 40-208	- Structural Bonding - Aluminum Honeycomb and Parts Cleaning.
NE 40-209	- Structural Bonding - Cleaning of Laminates and Non-Metallic Honeycomb Cores.
NE 40-230	- Automated Tape Laying (ATL) or Automated Fiber Placement (AFP) - Lamination Requirements.
NE 40-237	- Lamination Process by Resin Film Infusion (RFI) - Requirements.
NE 40-251	- Composite Parts, First Part Qualification (FPQ) - Procedures and Requirements.
NE 40-267	- Composite Material Equivalency and Processor - Equivalency Demonstration.
NE 42-012	- Raw Materials/Equipment/Parts Protection for Handling, Transportation, Picking, Feed Line and Storage.
NE 42-020	- Material and Equipment (Purchased and Manufactured) - Storage Conditions, Shelf Life and Revalidation Requirements.
NE 42-022	- Prepreg, Dry Fabric with Binder, Adhesive Film, Adhesive Foam, Non-Adhering Film and Adhesive Primer - Transportation, Handling, Storage and Revalidation.

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- NE 42-027 - Parts - Temporary Protection.
 NE 80-060 - Electrical Bonding - Repair for Composite Materials Parts with Expanded Copper Foil.
 PF 15-018 - Polymethacrylimide Rigid Foam - Cutting, Cleaning, Drying, Thermoforming, Heat Treatment and Machining.

Foreign Standards

SAE-AMS - Aerospace Material Specification
 AMS-C-9084 - Cloth, Glass, Finished, For Resin Laminates.

SAE-AS - Aerospace Standard
 AS9102 - Aerospace First Article Inspection Requirements.

4. DEFINITIONS

4.1 ABORT TEMPERATURE

Limit temperature a cure cycle may be aborted without losing material or part being cured.

4.2 AIR ENTRAPMENT

Air entrapped during the material deposition stage of prepreg layup that induces void or porosity in the cured laminate. It does not include parts' surface conditions.

4.2.1 Void

Air, water vapor, or gas that has been trapped and cured into a laminate or adhesive bond line.

4.2.2 Flat Porosity

An internal condition of composite structure which consists of small voids entrapped in a single ply or in the interface between two plies of the laminate.
 It is not a delamination, a bridging nor surface conditions.

4.2.3 Delamination

Separation of the layers in a laminate.

4.2.4 Bridging

Condition in which fibers do not move into or conform to negative radii and corners during molding, resulting in voids and dimensional control problems.

4.3 BAGSIDE

The side of the part cured facing the vacuum bag.

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4.4 BATCH

Quantity of material produced essentially in the same process or during one continuous process and having identical characteristics throughout, using a well-defined collection of material with a total commonality of raw materials and process history. This definition is applicable to fiber, fabric, resin, and adhesive materials.

Prepreg material batch is a production run of prepreg material that is pre-impregnated using one batch of resin and fabric, all in one continuous operation. A batch of prepreg may consist of one or more rolls of material.

NOTE

In case more than one batch of resin and/or fabric is used for prepreg batch production, it shall be accorded between Embraer and Manufacturer during Qualification and registered in the PROCESS CONTROL DOCUMENT (PCD) of the respective material.

4.5 BLEEDER

A nonstructural layer of material used in the manufacturing of composite parts to allow the escape of excess gas and resin during cure.

4.6 BREATHER

A permeable material that serves as a continuous vacuum path over a part.

4.7 BUTT JOINT

A type of edge joint in which the edge faces of the two parts of the same layer are a right angle to the other faces of the adherend.

4.8 OVERLAP JOINT

Joining of two parts of the same layer in which the surface of one part extends past the leading edge of another.

4.9 CAUL PLATE

Smooth plates (metal, plastic, rubber, carbon fiber, fiber glass composites, etc.) free of surface defects at the appropriate size and shape as the composite lay-up, used immediately in contact with the lay-up during the curing process to transmit normal pressure and provide a smooth surface on the finished part.

4.10 TOOLING (OR MOLD)

The molds and fixtures used to produce a composite article. Tooling may be made of any suitable material, including composites.

4.11 CLEAN ROOM

It is an area with controlled environmental conditions (temperature, humidity, and particles level) with positive pressure where the composite materials are cut, laid-up on tolling and vacuum bagged. Usual materials in this area are fiber glass, aramid and/or carbon fiber prepregs and adhesive films.

4.12 CO-CURING

The act of curing a composite laminate and simultaneously bonding it to some other uncured structure.

4.13 COMPOSITE MATERIALS

A combination of two or more materials (reinforcing elements, fillers, and composite matrix binder), differing in form or composition on a macro-scale. The constituents retain their identities; that is, they do not dissolve or merge completely into one another although they act in concert. Normally, the components may be physically identified and exhibit an interface between one another. In Polymeric Composite Materials, the reinforcements are fiber glass, aramid or carbon fibers and the matrixes are epoxy, phenolic, polyimide, or thermoplastic resins.

4.14 CORE

Lightweight material used between two skins to manufacture sandwich parts, often substantially thicker than the other layers. Core is usually honeycomb or foam.

4.15 CURE CYCLE

The schedule of time periods at specified conditions (heat, pressure and/or vacuum), to which a reacting thermosetting material is subjected in order to reach a specified property and permanently change a material system as a result of a controlled chemical reaction.

4.16 DWELL TEMPERATURE

An intermediate temperature in the cure cycle when a laminated part is kept at the same temperature to allow volatiles or entrapped air scaping from the laminated material and/or to permit resin staging/or gelling at a lower temperature than the curing temperature to prevent exothermic reaction during the subsequent heating up.

4.17 DWELL TIME

The period during which the temperature is maintained stable at the dwell temperature.

4.18 FABRIC

A generic material construction consisting of interlaced yarns of fibers, usually a planar structure. Specifically, a cloth woven in an established weave pattern from advanced fiber yarns and used as the fibrous constituent in an advanced composite lamina.

4.19 FABRIC DISTORTION

A deviation of a fabric yarn from a straight line, defined by the maximum yarn deviation along the yarn over which the deviation takes place.

4.20 WARP

The yarn running lengthwise in a woven fabric that is parallel to the selvage. A group of yarns in long lengths and approximately parallel.

4.21 FAYING SURFACES

Surfaces of cured parts that will be mated to other structures in a subsequent assembly.

4.22 FILL (OR WEFT)

The yarns running perpendicular to the longer dimension or selvage of the fabric reinforcement.

4.23 LAGGING THERMOCOUPLE

The thermocouple which indicates the slowest heat-up and cool-down rates.

4.24 LEADING THERMOCOUPLE

The thermocouple which indicates the fastest heat-up and cool-down rates.

4.25 MATRIX RESIN

The organic resin which encapsulates the reinforcement to produce the composite.

4.26 OUT TIME

It is the continuous or accumulated period of time between the material is first exposed to temperatures above storage condition (according to applicable MEP) and the start of cure cycle, in which the material maintains its handling, curing and mechanical properties. This period of time is the sum of all exposure events including, but not limited to, transportation, storage and shop floor (clean room). This definition does not specify tack time nor stand by time, only the total exposure time (Out time).

4.27 PART THERMOCOUPLE

A thermocouple placed in actual part, in vacuum bag during cure.

4.28 PARTING AGENT

Release material applied on the tooling to allow cured parts removal from the tool, including solid parting films and mold release agents.

4.29 COMPACTION (DEBULK)

The application of a temporary vacuum or pressure on the laminate to remove entrapped air and compact the layers during parts lamination.

4.30 PREPREG

Composite materials ready to mold or cure, in sheet form which may be tow, tape, cloth, or mat impregnated with resin (partially cured).

4.31 CRITICALLY LEVELS AND QUALITY CLASSES

Composite parts classification according to structural criticality levels and quality classes requirements that shall be specified in parts drawings, as described below:

4.31.1 Criticality Levels

a) LEVEL I - corresponds to parts (loads/structures/elements) which are essential to maintain the overall flight safety of the aircraft;

b) LEVEL II - corresponds to parts (structures/elements) that are not essential to maintain the overall flight safety of the aircraft, but its replacement is cost-critical; and

c) LEVEL III - corresponds to parts (structures/elements) that are not essential to maintain the overall flight safety of the aircraft and its replacement causes minor inconvenience and are not cost-critical.

4.31.2 Quality Classes

The quality classes refer to the maximum acceptable defect sizes by discrepancy type as stated in Tables 7 and 8.

4.32 SHELF LIFE

For this definition, see NE 42-022.

4.33 STEP CURING

An intermediate curing process used to fabricate any detail that will be later incorporated into a laminated part. This process is usually used to process sandwich structures in which a prepreg skin is molded (pre-cured) in a process separate from the core and subsequently bonded to the core.

4.34 SOAK TIME

The minimum time necessary to achieve parts' final curing (polymerization of the resin) at a specific temperature.

4.35 TAPE

Designation in which all fibers are aligned in parallel (in the same direction).

4.36 TOOLING (JIGS) FAMILY FOR QUALIFICATION

It is considered a family, all tooling with the same manufacturing concept, same material and with similar thermal behavior (thermal lag).

4.37 MODEL BASED DEFINITION (MBD)

Illustrative 3D drawing that captures every detail of products, projects information and engineering requirements.

4.38 FASTENER

Hardware device that mechanically joins or affixes two or more objects together (ex: rivet, pin, screw, etc.).

4.39 REWORK

Actions performed on a nonconforming product/part that, when completed, brings it into conformance with its original requirements.

4.40 REPAIR

Actions performed on a nonconforming product/part that, when completed, make it acceptable for its intended use/application.

4.41 WORK ORDER

Instructions that guide production teams through the step-by-step execution of manufacturing tasks, outlining all the necessary information to complete the task successfully. Also known as Job Order, Job Sheet, Production Order (“Ordem de Produção/OP”).

5. REQUIREMENTS

When there are conflicts between engineering drawing and this Standard, engineering drawing is mandatory. If there are conflicts between the Work Order and the engineering drawing or this Standard, the engineering drawing and this Standard are mandatory.

5.1 REQUIREMENTS FOR MATERIALS

Materials used in Embraer parts shall meet the requirements specified in one of the specification listed in item 3. Materials used in the fabrication process shall be qualified and approved by Embraer Materials Engineering prior to use.

5.2 REQUIREMENTS FOR THE FACILITY EQUIPMENT

All equipment used to prepare and control the parts shall be qualified by Quality Engineering.

5.2.1 Autoclave

a) it shall be used autoclave with heated air or inert gas circulation, with capability to provide a vacuum of at least 23 inHg, temperature range from room temperature to 121°C or 177°C and pressure range between 0 to 100 psi.

The autoclave shall meet the heat-up and cool down rate, dwell time, pressure and vacuum requirements specified in the specific Fabrication Process (PF's)/Engineering Standards (NE's) (listed in item 3) called out in the engineering drawings;

b) a qualified autoclave Class 5 or less shall be used, with a pyrometer per NE 40-077. The autoclave shall be qualified and controlled by Embraer Quality Control;

c) the autoclave shall have a continuous record of the vacuum, pressure and temperature measurements since the start-up, heat-up phase, dwell time, soak time and cooling down of the part up to the ambient temperature; and

d) the autoclave shall be maintained. A documented maintenance plan shall be validated by the Quality Engineering prior to its application, providing providences for (but it is not restricted to) identification of critical items and parts to be checked (e.g., plenum chamber, thermocouple contacts etc.), cleaning frequency and revalidation after retrofitting or replacement of critical elements (e.g., electrical resistance, compressors etc.). Records of all these operations are mandatory.

NOTE

The maintenance plan, validation and revalidation operations shall be compliant to engineering, quality and statutory requirements applicable.

5.2.2 Oven

a) it shall be used an oven with heated air circulation, vacuum capability of at least 23 inHg, and minimum temperature of 177°C (350°F). The ovens shall be capable to provide the heat-up and cool down rate, dwell time, soak time and vacuum requirements of the specific Fabrication Process (PF's)/Engineering Standards (NE's) (listed in item 3) called out in the engineering drawings;

b) a qualified oven Class 5 or less shall be used, with a pyrometry per NE 40-077. The ovens shall be qualified and controlled by Embraer Quality Control;

c) the oven shall have a continuous record of the vacuum and temperature measurements since the start-up, heat-up phase, dwell time, soak time and cooling down of the part up to the ambient temperature; and

d) the oven shall be maintained. A documented maintenance plan shall be validated by the Quality Engineering prior to its application, providing providences for (but it is not restricted to) identification of critical items and parts to be checked (e.g., plenum chamber, thermocouple contacts etc.), cleaning frequency and revalidation after retrofitting or replacement of critical elements (e.g., electrical resistance, compressors etc.).

Records of all these operations are mandatory.

NOTE

The maintenance plan, validation and revalidation operations shall be compliant to engineering, quality and statutory requirements applicable.

5.2.3 Clean Room

a) the lay-up room to process pre-impregnated materials, honeycombs, rigid foams and adhesives, shall meet the requirements of NE 07-002;

b) the materials or procedures listed below are prohibited in the lay-up area:

1 - Uncured parting agents.

2 - Hand creams.

3 - Application of spray coatings (except tackifier) or use of aerosol dispensers, except cleaning with water based commercial cleaners (cover any prepreg or uncured adhesive film within 3 m).

4 - Use of waxes, compounds containing uncured silicone or any material detrimental to adhesion, such as dirt, grease or oil.

5 - Compressed air that has visible signs of oil, moisture or solid particles.

6 - Processes or operations which produce uncontrolled spray, dust, fumes or particulates.

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c) the following procedures are allowed in the limited contamination area:

- 1 - Cleaning materials of cured composite surfaces in accordance with NE 07-002 is allowed.
- 2 - Peel ply removal from cured parts shall be at least 1 m away from any prepreg or uncured film adhesive. All particles shall be vacuumed with a vacuum system or solvent wiped.
- 3 - Inkjet marking.
- 4 - Application of tackifier solution to tool surfaces, take care should be taken to minimize the overspray. Material application shall be regulated by Occupational Safety Department.
- 5 - Spraying of water based commercial glass cleaner for cleaning.

5.2.4 Freezer

The freezer shall be capable to supply the maximum desired temperature to storage the perishable materials, usually -18°C (0°F) during all the time. Spikes in temperature are allowed during frozen storage. The maximum temperature reading of a spike shall not exceed $-6,6^{\circ}\text{C}$ (20°F). Time period above the specified storage temperature shall not exceeds 15 minutes, with a total time for all periods not to exceed 60 minutes in any 24 h period.

Additional exposure units do not have to be added for temperature spikes within these limits:

a) the freezer shall preferably have continuous temperature record or shall record the temperature at least every 15 minutes. If the temperature falls outside the specified limits in the raw material specification, the temperature shall be recorded just when falls outside the specified limits and when returning within the acceptable limits;

b) adhesive films, prepregs, adhesive foams, and others perishable materials shall be cold temperature storage (in freezers), in their original closed containers until use and may only be opened, handled and room temperature stored within the environmentally controlled area;

c) on receiving, all rolls of perishable prepreg materials and adhesive films shall be cold temperature stored and the storage life shall be according to NE 42-022; and

d) the freezer shall be maintained. A documented maintenance plan shall be validated by the Quality Engineering prior to its application, providing providences for (but it is not restricted to) identification of critical items and parts to be checked (e.g., condenser, compressor, thermostat, thermocouple contacts etc.), cleaning frequency and revalidation after retrofitting or replacement of critical elements (e.g., thermostat, evaporator, compressors etc.).

Records of all these operations are mandatory.

NOTE

The maintenance plan, validation and revalidation operations shall be compliant to engineering, quality and statutory requirements applicable.

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5.2.5 Automated Tape Laying (ATL) and Automated Fiber Placement (AFP) Machines

Automated tape laying (ATL) and automated fiber placement (AFP) machines are used to increase rate and precision in the production of advanced composite parts on molds (tools) to form automated prepreg layups according to NE 40-230.

5.2.6 Heat Guns and Tacking Irons

To facilitate material layup, heat gun and/or tacking iron devices may be used for local heating, respecting:

1. heating devices shall NOT contact tooling surfaces nor composites layers;
2. tooling surfaces and composite layers shall NOT exceed the temperature of 65°C; and
3. heating devices shall be in continuous movement to prevent local over heating.

5.2.7 Tool Leak Check

Tooling leakage check shall be performed before releasing it to production area. Tooling leakage is not allowed therefore it shall be continuously monitored for leaks.

5.3 REQUIREMENTS FOR MANUFACTURING

5.3.1 Transportation, Handling and Storage

For aircraft category items, see NE 42-012.

For transportation, handling, packing and storage, also consult NE 42-020, NE 42-022 and NE 42-027.

5.3.2 Perishable Materials

For transportation, handling, storage and revalidation, see NE 42-022.

5.3.3 Core Materials

Non-perishable materials, such as honeycomb core, rigid foam and others, shall be stored preferably in their original containers or, when necessary, in dust proof plastic film such as nylon or polyethylene.

- a) Aluminum honeycomb core shall be cleaned, handled and stored according to NE 40-208.
- b) Polyimide honeycomb core shall be cleaned, handled and stored according to NE 40-209.
- c) Polymethacrylimide rigid foams shall be cleaned, handled and stored according to PF 15-018.

5.3.4 Tooling (or Molds)

Tooling Manufacturing Engineering is responsible for tooling qualification. Only qualified tooling shall be used in composite parts manufacturing.

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5.3.5 Layup

The layup shall be performed inside the clean room by hand layup or automated processes, following the stacking sequence specified in the drawings and/or in the Work Order or another equivalent document.

The prepreg materials, adhesive films and adhesive foams shall be handled according to NE 42-022. Non-metallic honeycomb (MEP 15-010) shall be handled according to NE 40-209.

- a) personnel involved in composite parts lay-up process shall be qualified according to special process production requirements and certified by Quality Engineering;
- b) prepreg materials with different resin (except for adhesive film) are not allowed to be laminated in the same part, unless specified in engineering drawing;
- c) all manufactured parts shall be controlled and monitored, ensuring raw material and process traceability;
- d) it is NOT allowed to incorporate the selvage (including the sewing line) of woven prepreg materials into the part (see Figure 1);

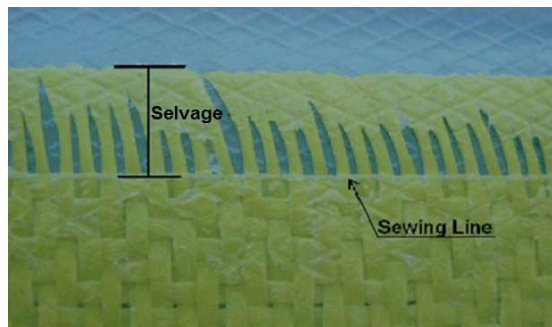


FIGURE 1 : SELVAGE REGION

- e) splice, positioning and direction of plies layup requirements are listed in Table 2;
- f) reject any uncured kit (prepreg and/or adhesive film, foam) or parts suspected of condensed moisture;
- g) record information about autoclave load(s), run number(s), thermocouples and tooling identification;
- h) it is allowed to apply tackifier to promote tack between the first material ply (e.g. prepreg, adhesive film) to the mold surface;
- i) tackifier may be used to join various bagging materials, including, but not limited to, breather and bleeder, when necessary, but any application between prepreg layers or prepreg and adhesive film is prohibited; and

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j) whenever uncured prepreg parts lamination on the molds have not been finished in the same shift and the lamination activity is not going to be continued in the next hour, the unfinished parts shall be protected from contamination by covering them with clean polyethylene or nylon film.

5.3.5.1 Traceability Information

Each individual layer shall be identified with the following information: part number, ply orientation and ply number stacking sequence. There shall be evidence ensuring that the layup sequence is as specified in drawings.

The layers may be stored in kits, identified with part number and number of plies before sealing the kit bag. Kits may be stored in the freezer.

a) each sealed kit shall correspond to a single part. When mixed materials in a precut present conflicting storage temperature and storage life requirements, the most restrictive requirements shall be used, i.e., lowest temperature, shortest life, etc.

Table 1 presents ply control methods.

5.3.5.1.1 Automated Lay-up Process

Automated Tape Laying (ATL) and Automated Fiber Placement (AFP): the evidence may be provided by machine data acquisition system. It may be stored in data system or printed and included in the Work Order document traceability information.

It is necessary to ensure traceability of lay-up program to the Work Order document.

5.3.5.1.2 Hand Lay-up Processes

a) electronic oriented lay-up process (Laser Projection System):

- Cutting operation: paper tag or ink mark identification; and

- Lay-up operation: the operator shall follow and check according to laser projection information. It is necessary to ensure traceability of the laser projection program to the Work Order document.

b) operator manually orientation lay-up process:

- Cutting operation: paper tag or ink mark identification;

- Lay-up operation: electronic data acquisition system for ply information.

It may be stored in data system or printed and included in the Work Order document traceability information or physical evidence through label transfer from the plies Work Order document as they are laminated;

c) the prepreg kit shall be identified with the information specified in the item 5.3.5.1 and the layers sequence shall be inspected during the lay-up and before being sealed to be stored in the freezer.

TABLE 1 : LAY-UP PLY CONTROL METHODS

LAY-UP PROCESS	PLY POSITIONING AND LAY-UP METHOD	PLY IDENTIFICATION METHOD IN THE CUTTING OPERATION	METHOD OF ASSURANCE OF LAY-UP SEQUENCE DURING LAY-UP	OBSERVATION
AUTOMATED - ATL	NUMERIC CONTROL	NOT APPLICABLE	NUMERIC CONTROL LAY-UP	IT IS NECESSARY TO ENSURE TRACEABILITY OF LAY-UP PROGRAM IN THE WORK ORDER DOCUMENT
AUTOMATED - AFP	NUMERIC CONTROL	NOT APPLICABLE	NUMERIC CONTROL LAY-UP	IT IS NECESSARY TO ENSURE TRACEABILITY OF LAY-UP PROGRAM IN THE WORK ORDER DOCUMENT
HAND LAY-UP - (HDLUP)	LASER PROJECTION ORIENTED	INK MARK AND/OR PAPER TAG	THE OPERATOR SHALL FOLLOW AND CHECK ACCORDING TO LASER PROJECTION INFORMATION	IT IS NECESSARY TO ENSURE TRACEABILITY OF THE LASER PROJECTION PROGRAM TO THE WORK ORDER DOCUMENT
	ORIENTED MANUALLY BY OPERATOR	INK MARK	ELECTRONIC DATA ACQUISITION SYSTEM FOR PLY INFORMATION	IT MAY BE STORED IN DATA SYSTEM OR PRINTED AND INCLUDED IN THE WORK ORDER DOCUMENT FOR TRACEABILITY INFORMATION
		PAPER TAG	PHYSICAL EVIDENCE THROUGH LABEL TRANSFER FROM THE PLIES TO WORK ORDER DOCUMENT	THE EVIDENCE SHALL BE DONE AS THE PLIES ARE BEING LAMINATED

5.3.5.2 Control Coupon Requirements

All parts level I requires one control coupon (witnessing coupon) for the main raw material used in laminate. Materials utilized to electrical bonding and surface finishing (for example first ply and last ply, Tedlar®, cooper foil, glass) are not applicable. In case of small and/or simple parts, materials engineering shall decide the need of control coupons. In case of small and/or simple parts, materials engineering shall decide the need of control coupons. Other part levels shall have control coupons only if specified in the drawing or manufacturing document.

The process control coupons shall be cured inside the same vacuum bag of the part. Whenever a process control coupon is required, the mold design shall provide an adequate room for it.

The number of plies, orientation and layup of control coupons panel shall be as specified in the material specification (MEP) for interlaminar shear strength coupons. Control coupons panel dimensions shall be according to Figure 2.

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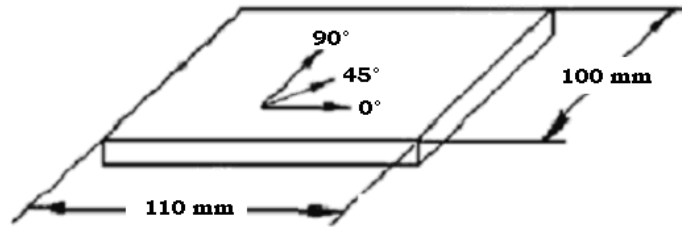


FIGURE 2 : CONTROL PANEL

Once the control coupons panel is cured, a minimum of 5 samples, in 0° direction, shall be cut.

The dimensions of each one, the test method and the acceptance criteria are specified in each material respective MEP.

NOTE

Control coupons (samples) shall be tested only if there is any discrepancy during the cure cycle and the results shall be evaluated by Embraer Materials Engineering.

5.3.5.3 Splice Requirement

Splices (overlap joint or butt joint) in the same direction shall be staggered by 25 mm minimum. In addition, splices aligned through the part/laminate thickness (coincident splices) shall be separated by at least four plies, regardless of the ply angle orientation. See Figures 3 and 4 for details.

NOTES

- 1) Splices may cross one another in the stacking sequence.
- 2) Different splices are acceptable if specified in the drawing.
- 3) ATL and AFP processes requirements shall be according to NE 40-230.

5.3.5.4 Overlap Joints

Where overlap joints are permitted, the minimum overlapping dimension shall be 5 to 10 mm, unless otherwise specified on the drawing. See Figures 3 and 4 and Table 2.

NOTE

For non-structural, galvanic corrosion protection, fiberglass layer, item 5.3.5.4 is not applicable, and the overlap may be different from the pattern herein stated.

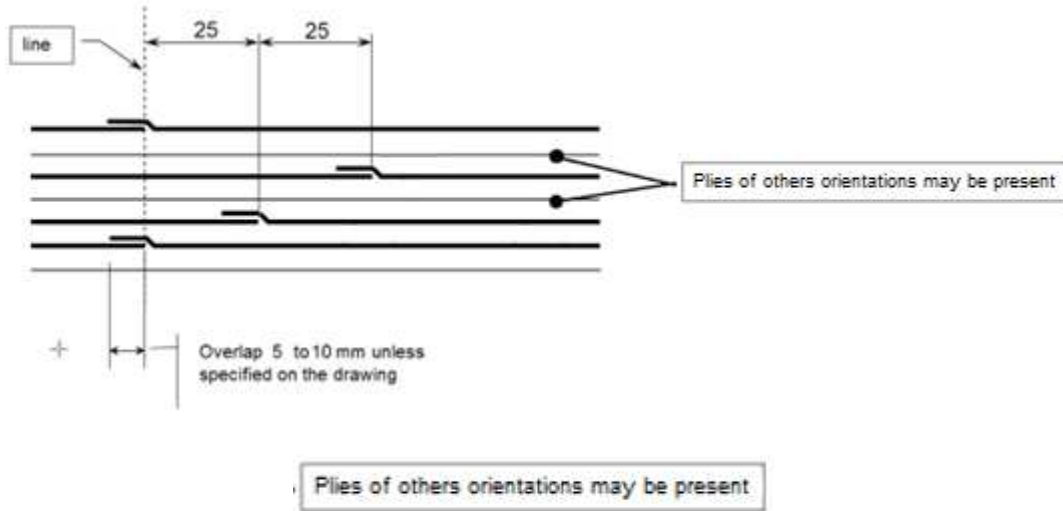


FIGURE 3 : OVERLAP JOINTS (SPLICE) DIMENSIONS

Overlap joints of Tedlar®, electrical bonding mesh (pre-impregnated or dry mesh) or shall be from 15 to 25 mm (Figure 4).

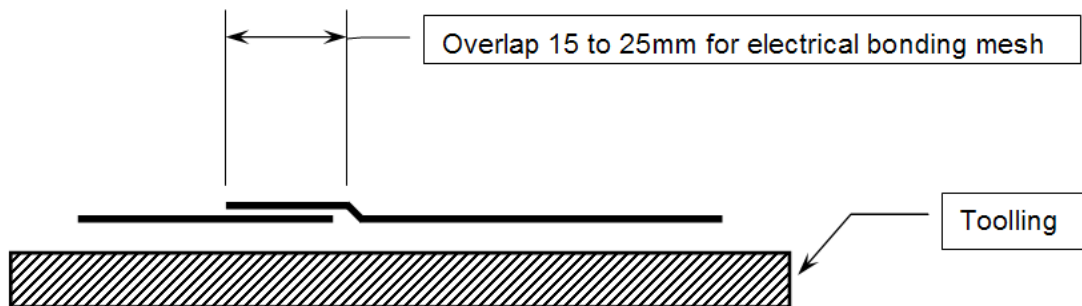


FIGURE 4 : OVERLAP SPLICE (TEDLAR®, WIRE MESH AND FOILS) DIMENSIONS

5.3.5.5 Butt Joints

If not specified in drawing, the ply to be joined by butt-joint (spliced) technique may be a continuous ply and the gap between two plies shall be 0 to 2 mm. See Figure 5 and Table 2.

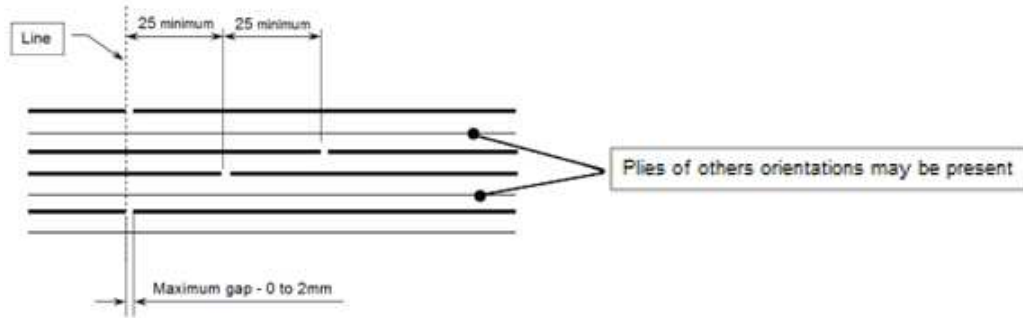


FIGURE 5 : BUTT JOINT (SPLICE) DIMENSION

TABLE 2 : SPLICE REQUIREMENTS

MATERIAL		DESCRIPTION	SPLICE REQUIREMENTS			LAY-UP	
			STAGGER (MIN)	OVERLAP	GAP (MAX)	POSITION	DIRECTION
FABRIC	CARBON	FAYING REGION	25 mm	NOT ALLOWED	2 mm	±5 mm OF LOCATION SPECIFIED ON DRAWING	±5 DEGREE
		NONFAYING REGION	25 mm	5 TO 10 mm	NOT ALLOWED		
	GLASS	FAYING REGION	25 mm	NOT ALLOWED	2 mm		
		NONFAYING REGION	25 mm	5 TO 10 mm	NOT ALLOWED		
	KEVLAR®	FAYING REGION	25 mm	NOT ALLOWED	2 mm		
		NONFAYING REGION	25 mm	5 TO 10 mm	NOT ALLOWED		
TAPE (HAND LAY-UP PROCESS) (1)	CARBON	FAYING REGION	25 mm	NOT ALLOWED	2 mm		±2 DEGREE
		NONFAYING REGION	25 mm	NOT ALLOWED	2 mm		
SURFACE FILM	TEDLAR®	---	N.A	15 TO 25 mm	NOT ALLOWED	N.A	N.A
	ADHESIVE FILM	---		MAX 15 mm	2 mm		
	COPPER MESH	---		15 TO 25 mm	NOT ALLOWED		
NON-STRUCTURAL LAYERS (EX: GALVANIC CORROSION PROTECTION) (2)	FIBERGLASS FABRIC	---	N.A	MAX 25 mm	NOT ALLOWED	N.A	N.A

(1) For automated laying processes (ATL and AFP), see NE 40-230.

(2) Not allowed changing faying surfaces, steps and fasteners joint.

N.A= Not Applicable.

5.3.6 Requirements for Honeycomb Before Lay-up

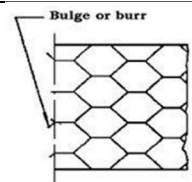
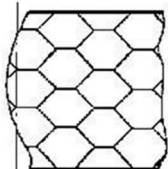
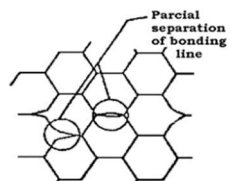
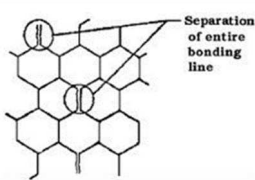
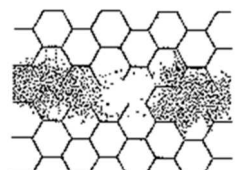
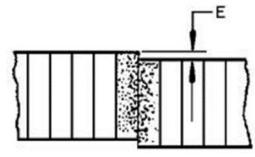
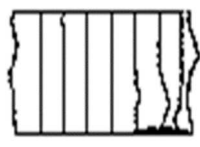
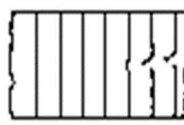
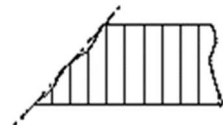
The honeycomb, before starting lay-up, shall meet the requirements of contour machining cutout according to NE 03-047 and Table 3, where the acceptance limits and the corrective procedures are given.

TABLE 3 : DEFECTS LIMITS FOR HONEYCOMB CORE

DEFECTS	ACCEPTANCE LIMITS	CORRECTION LIMITS	CORRECTIVE PROCEDURES
FRAYING OR BURRS	SHALL NOT BE GREATER THAN 2,5 mm	SINCE THE FINAL DIMENSIONS OF THE CORE ARE NOT AFFECTED	USE SAND PAPER OR FINE ABRASIVE TO ELIMINATE THE DEFECT
WAVY EDGE	ALL SIZES	MRB DISPOSITION	MRB DISPOSITION
PARTIAL NODE DISBONDING (LESS THAN HALF OF BONDED AREA)	MAX 34% OF NODES WITHIN ANY AREAS OF 18 cm ² LIMITED TO MAX OF 5% IN THE PART	MRB DISPOSITION	MRB DISPOSITION
TOTAL NODE DISBONDING (MORE THAN HALF OF BONDED AREA)	MAX 2 NODES WITHIN ANY AREA OF 13 cm ² LIMITED TO MAX OF 1% IN THE PART	MRB DISPOSITION	MRB DISPOSITION
DISBONDING BETWEEN SPLICED CORES	IT SHALL BE A CONTINUOUS BONDING OR JOINING HAVING 5% FAILURE AT MOST IN JOINT.	CORRECTION AT BONDING LIMITED TO 20% OF JOINT OF CORES. LARGE DEFECTS SHALL BE REMOVED AND RE-BONDED	REPAIR BONDING USING RESIN WITH MICROSPHERES OR CUT OFF ALL JOINT AND REPEAT THE PROCESS
SECTIONED HONEYCOMB	NONE	NONE	JOIN NEW CORE AND BOND
RIPPLE IN CORE CHANFER	ALL SIZES	MRB DISPOSITION	MRB DISPOSITION

The Table 3.1 shows the visual aspects of defects for honeycomb core before lay-up.

TABLE 3.1 : VISUAL ASPECTS OF DEFECTS FOR HONEYCOMB CORE

DEFECT	DEFINITION	VISUAL ASPECT
A - PERIPHERY (1) FRAYING OR AREAS WITH BURRS	ANY MATERIAL, BULGE OR BURR BEYOND SECTION LINE	
(2) WAVY EDGE	ANY DEVIATION OF EDGE SURFACE BEYOND THAT SPECIFIED ON THE PRODUCT DRAWING	
B - NON-PERIPHERIC (1) PARTIAL DISBONDING OF HONEYCOMB	PARTIAL DISBONDING LIMITED TO HALF OF THE HONEYCOMB CELL TAPE SURFACE	
(2) TOTAL DISBONDING OF HONEYCOMB	DISBONDING OF MORE THAN HALF OF THE HONEYCOMB CELL TAPE BONDING	
C - CORE SPLICING (1) DISPLACEMENT OF THE CORES	ANY DISCONTINUITY IN THE BONDING LINE BETWEEN THE HONEYCOMB CORES JOINED BY FOAM OR INTERLOCKING	
(2) DIFFERENT THICKNESS OR STEPS	DIFFERENCE IN THICKNESS OR IN THE ALIGNMENT BETWEEN JOINED CORES	
D - BUCKLING LINES	THESE ARE DEFORMATION LINES IN CELL WALLS CAUSING A DEPRESSION IN CORE	
E - CUT CELLS	ANY CUT IN THE HONEYCOMB CELL WALL	
F - RIPPLE IN CORE CHAMFER	DIFFERENCES BETWEEN THE PRODUCT DRAWING AND THE SURFACE OF CORE CHAMFER	

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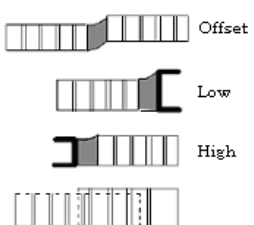
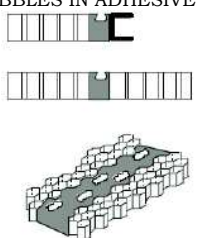
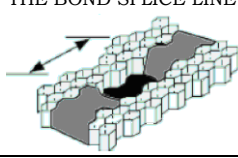
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5.3.7 Requirements for Spliced Honeycomb Core

The use of adhesive foam between spliced cores shall be indicated by the Manufacturing Engineering except when:

- Adhesive foam application is required in drawing;
- After splicing the cores, the part weight does not exceed the drawing weight requirement and the weight is under the acceptable limits described in Table 4; and
- For interior parts, the materials used for splicing the cores shall meet the applicable flammability requirements.

TABLE 4 : DISCREPANCY ACCEPTANCE CRITERIA FOR SPLICED HONEYCOMB CORE

DISCREPANCY	ACCEPTANCE LIMITS
A. THICKNESS MISMATCH IN BONDLINES 	SHALL NOT EXCEED 0,5 mm.
B. BUBBLES IN ADHESIVE FOAM 	SURFACE BUBBLES SHALL NOT EXCEED ONE CORE CELL SIZE IN THE LONGEST SURFACE DIMENSION.
C. BONDLINE FULL DEPTH GAP ACROSS THE SPLICE  DISCREPANCY LENGTH – THE DIMENSION PARALLEL TO THE BOND SPLICE LINE. 	THE MAXIMUM PERMISSIBLE LENGTH FOR GAP DEPTH (EMPTY CELLS) IS 13 mm, OR THREE (3) ADJACENT CELLS, WHICHEVER IS SMALLER, PER EACH LINEAR 300 mm AND THIS GAP SHALL NOT EXCEED 5% OF THE RATE: $\frac{\text{HONEYCOMB DEFECT AREA}}{\text{HONEYCOMB BOND AREA (1)}}$ (1) THE HONEYCOMB BOND AREA IS EQUAL TO THE SPLICE HEIGHT TIMES THE SPLICE LENGTH.
D. EXCESSIVE ADHESIVE	MAXIMUM SIZE OF TWO (2) CELLS (PERPENDICULAR TO THE BONDING LINE) OR 9,5 mm (WHICHEVER IS LARGER) BEYOND THE CORE EDGE OF A SPLICE OR BEYOND THE DESIGNATED AREA TO BE POTTED

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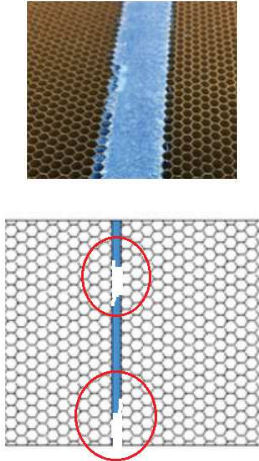
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TABLE 4 : DISCREPANCY ACCEPTANCE CRITERIA FOR SPLICED HONEYCOMB CORE

E. DISBONDING BETWEEN SPLICED CORES  UPPER: spliced cores with adequate foam adhesive; BOTTOM: core splices disbonded (may have residual foam)	UP TO 5% FAILURE: DISCONTINUITY OF BONDED CORES ARE ACCEPTED 5% TO 20% FAILURE: REPAIR BONDING USING RESIN WITH MICROSPHERES (MEP 10-051) OR CUT OFF ALL JOINT AND REPEAT THE PROCESS LARGER THAN 20% OF THE BONDED AREA REQUIRES MRB DISPOSITION DISBONDING UP TO 5% OF BONDED AREA (SPLICED CORES LENGTH TIMES HEIGHT) IS ACCEPTED DISBONDING FROM 5% TO 20% OF BONDED AREA: REPAIR AREA USING RESIN WITH MICROSPHERES (MEP 10-051) OR CUT OFF ALL JOINT AND REPEAT THE PROCESS DISBONDING LARGER THAN 20% OF THE BONDED AREA REQUIRES MRB DISPOSITION
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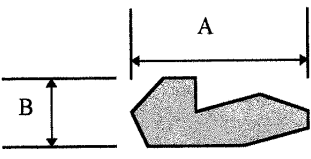
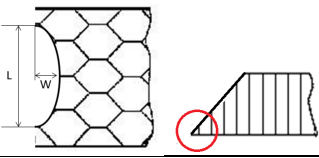
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5.3.8 Requirements for Complete Stabilized Honeycomb Core

The completely stabilized honeycomb core (according to NE 40-188) shall meet the requirements of contour cutout according to NE 03-047 and Table 5 where the acceptance limits and the corrective procedures are given. The visual inspection method shall be according to NE 07-009.

TABLE 5 : ACCEPTANCE CRITERIA FOR STABILIZED HONEYCOMB

DEFECTS	DEFINITION	ACCEPTANCE LIMITS	CORRECTIVE PROCEDURES
EXPOSED CORE DUE TO MACHINE PROCESS (CHAMFER) EXPOSED CORE AT THE CONTOUR (NO CORE DAMAGE OR CORE MISSING)	PEEL PLY FAILURE OR ANY OTHER DISCREPANCY EXTENDING FROM CORE AREA TO AN AREA EXPOSED TO THE ENVIRONMENT PEEL PLY MISSING OR ANY OTHER DISCREPANCY THAT EXPOSE THE CORE TO THE ENVIRONMENT (GENERALLY DUE TO CHAMFER CUTTING MACHINE ENTRANCE) 	NOT ACCEPTABLE FOR ITEMS TRANSFERRED BETWEEN DIFFERENT LAMINATION SITES	INDIVIDUAL DISCREPANCIES, ALL CLASSES: $A + B \leq 100 \text{ mm}$ REPAIR AS FOLLOWS: CLEAN WHOLE AREA ACCORDING TO NE 40-209, APPLY POTTING MATERIAL (MEP 10-051) AND CURE AS PER MANUFACTURER INSTRUCTIONS BEYOND ACCEPTANCE LIMIT, MRB DISPOSITION
		FOR ITEMS HANDLED IN SEQUENCE WITHIN THE SAME LAMINATION SITE, THE MAXIMUM INDIVIDUAL HOLE DISCREPANCY ALLOWED FOR ALL CLASSES IS: $A + B \leq 100 \text{ mm}$	DURING LAMINATION, EACH DISCREPANCY SHALL BE COVERED WITH ORIGINAL ADHESIVE FILM (MAXIMUM 10 mm OVERLAP) BEYOND ACCEPTANCE LIMIT, MRB DISPOSITION
PEEL PLY BRIDGE IN THE CHAMFER CORNERS	USUALLY ON CHAMFER CORNERS, THERE IS A BRIDGE ON THE PEEL PLY (ADHESIVE FILM PLY IS CORRECTLY STABILIZED)	ACCEPTABLE BRIDGE WIDTH $\leq 30 \text{ mm}$	NO CORRECTIVE PROCEDURE, MRB DISPOSITION
BELLY AT CORE EDGE (BOTTOM CHAMFER)	ANY DEVIATION OF EDGE SURFACE BEYOND THAT SPECIFIED ON THE PRODUCT DRAWING 	INDIVIDUAL DISCREPANCIES: ALL CLASSES: LENGTH $\leq 50 \text{ mm}$ WIDTH $\leq 15 \text{ mm}$	NO CORRECTIVE PROCEDURE, MRB DISPOSITION

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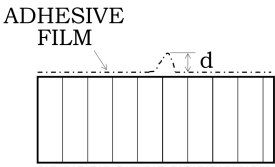
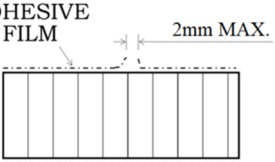
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TABLE 5 : ACCEPTANCE CRITERIA FOR STABILIZED HONEYCOMB

ADHESIVE FILM WRINKLE (NOT ACCEPTABLE TO CHAMFERED AREAS)	WRINKLE HEIGHT "Pd" SKETCH: 	CLASSES 1 AND 2: NONE ALLOWED CLASS 3: WRINKLE HEIGHT $\leq 0,4$ mm; WRINKLE LENGTH ≤ 70 mm, MAXIMUM ONE OCCURRENCE PER $0,5$ m² CLASS 4: WRINKLE HEIGHT $\leq 0,8$ mm; WRINKLE LENGTH ≤ 100 mm, MAXIMUM FOUR OCCURRENCES PER $0,5$ m² NONE ALLOWED ON FAYING SURFACES	REPAIR FOR CLASSES 3 AND 4: REMOVE (DO NOT SAND) THE WRINKLED TIP; REMOVE ANY LOOSE FLAKES - FOR ITEMS TRANSFERRED BETWEEN DIFFERENT LAMINATION SITES, IT IS ALLOWED A MAXIMUM REMAINING GAP OF 2 mm ADHESIVE FILM  2mm MAX. - FOR ITEMS HANDLED IN SEQUENCE WITHIN THE SAME LAMINATION SITE, IT IS ALLOWED A MAXIMUM REMAINING GAP OF 3 mm PROVIDED THAT IT IS COVERED WITH ORIGINAL ADHESIVE FILM (MAXIMUM 10 mm OVERLAP) DURING LAMINATION BEYOND ACCEPTANCE LIMIT, MRB DISPOSITION
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5.3.9 Preparation for Parts Curing

5.3.9.1 Thermocouple Positioning

For each curing load, at least 3 parts (with the highest thermal inertia) shall be used to monitor and control the cure cycle.

For each part with the highest thermal inertia, the controlling thermocouples shall be positioned at tooling areas with the highest thermal inertia, according to FPQ (part and tooling) thermal profiles (see item 6.2).

NOTES

1) If necessary, thermocouples shall be thermally isolated from the surrounding environment to measure only the tooling temperature.

2) For level I parts, at least one monitoring thermocouple shall be positioned at the lowest thermal inertia and one at the highest thermal inertia areas, as defined in FPQ thermal profiles (see item 6.2.1). It is recommended to use additional monitoring thermocouples in case of thermocouple failure during the cure.

5.3.9.2 Vacuum Bag Construction

5.3.9.2.1 All sharp edges, bolts, drilled holes, etc. that might puncture the vacuum bag shall be covered with tooling protection materials which, in turn, shall not be in direct contact with composite materials, e.g., prepreg, adhesive etc.

5.3.9.2.2 The vacuum bag construction, the number of bleeders, breathers, etc. shall be in accordance with the Embraer Fabrication Process (PF)/Engineering Standard (NE) documents.

5.3.9.2.3 Bag wrinkles are undesirable and every effort shall be made to eliminate them. To avoid rejection for wrinkles, air entrapment, voids, etc. the part may be left connected to an active vacuum source between completing the bagging operation and placing it in the autoclave for curing.

5.3.9.3 Leakage Test

The leakage test shall be performed following the procedures specified in the Fabrication Process (PF)/Engineering Standard (NE) documents for the material lay-up.

5.3.10 Curing Cycle

Laminated assemblies shall be cured according to the curing cycle specified in the Fabrication Process (PF)/Engineering Standard (NE) documents for the materials used in part manufacturing.

5.3.10.1 The curing cycle shall be fully respected as per each material NE standard. Any deviation in the curing cycle shall be registered in the Work Order and shall be evaluated by Quality and Materials Engineering; the engineering disposition shall be recorded in the Material Revision Board (MRB).

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The following exceptions are acceptable:

- 1) If the monitoring thermocouples are interrupted (temporary oscillation or permanent loss of recording data) during the cooling stage, parts Level I, II and III may be approved directly by Quality Engineering provided that other parts, with higher thermal inertia and within the same curing cycle as the affected parts, have accomplished the cure cycle specified in the Fabrication Process (PF) or Engineering Standard (NE).
- 2) If the monitoring thermocouples are interrupted (temporary oscillation or permanent loss of recording data) during any stage of the cure rather than the cooling stage. Parts levels II and III may be approved directly by Quality Engineering provided that other parts, with higher thermal inertia and within the same curing cycle as the affected parts, have accomplished the cure cycle specified in the Fabrication Process (PF) or Engineering Standard (NE).

NOTE

Only thermocouples connected to parts and/or molds are considered monitoring thermocouples; autoclave thermocouples shall NOT be used in the analyzes of the above exceptions.

5.3.11 Parts Inspection

5.3.11.1 Visual Inspection

The acceptance criteria for surface discontinuities shall be according to Table 7 and the inspection method according to NE 07-009. The visual inspection shall be applied to each production part.

5.3.11.2 Dimensional Inspection

The dimensions and tolerances of the finished parts shall be according to the specific drawing. General tolerances shall be according to NE 03-047.

NOTE

For MBD methodology, dimensions not presented in the production drawing shall be measured directly in the 3D model and for non-tolerated geometries, apply the general tolerances herein specified.

TABLE 6 : CURED PARTS NOMINAL THICKNESS PER PLY AND NOMINAL AREA WEIGHT

EMBRAER CODE	MATERIAL PART NUMBER	SPECIFICATION	NOMINAL WEIGHT g/m ²	NOMINAL THICKNESS (mm)
E1910379	116-50-CS550	AMS-C-9084	170 (1)	0,100
E1910363	7781-60-CS619		585 (2) (3)	0,350
E1453361	AF191XS .045	MEP 02-009	219	0,100 (6)
E4272241	3CU7-100 AN		195	0,100 (6)
E8056407	3CU7-125AN		142	0,100 (6)
E5921275	TWH10BS3	MEP 06-029	--- (9)	0,025
E3017832	TWH20BS3		--- (10)	0,050
E2086338	AF191M .03 WT	MEP 09-063	146	(4)
E2086261	AF163-2K .045 WT		220	(4)
E1910507	116-38-F155-116(L5926)	MEP 15-006	162	0,099
E4445043	L-530-7781		504 (3)	0,304
E1910496	7781-38-F155-117(L5765)		465	0,230
E3016816	K49-220-50-F155-114	MEP 15-007	165	0,100
E1910563	P1177		326	0,230
E1390060	116-38-F161-226	MEP 15-008	162	0,100
E1910513	P1178		424	0,210
E3015139	P1179	MEP 15-009	281	0,230
E9121495	T7D150-10-F161		280	0,160
E9122950	TXX-190-12-F263-29	MEP 15-014	288	0,190
E9123148	F3X-584-42-F263-28		552	0,350
E9125613	B021032	MEP 15-016	366 380 (3)	0,229
E3671097	P1180	MEP 15-018	552	0,370
E7725616	P1181E		324	0,210
E1967353	P0168		297	0,200
E2070720	P1140	MEP 15-022	544	0,350
E6606938	P0150		271	0,180
E2192960	P0160		288	0,180
E2787845	P1156		561	0,350
E2192954	P1157		561	0,350
E2061482	F4G584-42-F650	MEP 15-025	536	0,350
E8354083	HT93-120 (L5680)	MEP 15-026	191	0,130
E8353578	B021033		477	0,292
E8353584	HT93-49-285 (L5683)	MEP 15-027	318	0,263
E2127725	CPH2284/K49-120/38"		132	0,104
E5881360	CPH2284PB/KEVLAR 285		318	0,263
E8353623	B021035	MEP 15-028	566	0,400
E5881376	CPH2284PB/T300,3K,8HS		604	0,400
E2961345	P1142V10	MEP 15-047	322	0,210
E7724212	P1142E		322	0,210
E2957491	P0007V04		297	0,190
E1453386	P0007/A60B		297	0,190

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TABLE 6 : CURED PARTS NOMINAL THICKNESS PER PLY AND NOMINAL AREA WEIGHT

EMBRAER CODE	MATERIAL PART NUMBER	SPECIFICATION	NOMINAL WEIGHT g/m ²	NOMINAL THICKNESS (mm)
E8173290	P0017/A60C	MEP 15-054	288	0,192 (ATL & HDLUP)
E4593517	P0017/A30C			
E7781680	P0017/A32-3			
E7406985	P0017/A64-3		296	0,186 (AFP)
E9748355	S100367			
E1242020	111057F (FILM)	MEP 15-055 (11)	685	0,386
E1903943	MCX 123-1270	MEP 15-063 (11)		
E9499281	113536F	MEP 15-055	685	0,386
E2662764	MCX 1431270	MEP 15-063		
E5222256	K30163 (FABRIC)	MEP 15-057	639 (5)	0,374 (7) 0,393 (8)
E9025086	029858C (RESIN)	MEP 22-013		
E2340714	B021500	MEP 15-059	322	0,210
E3169880	P1165	MEP 15-060	296	0,240
E3152673	P0167		296	0,240
E702773	P0170		181	0,100
E3889379	B021545	MEP 15-064	TBD	0,213
E5294499	P2352W-19L-150	MEP 15-067	289	0,188
E4323506	FGF7781-29E-965	MEP 15-068	426	0,212
E1676818	FM6673G-37KL-965	MEP 15-070	TBD	0,220
E1186211	030301R	MEP 15-071	TBD	0,240

ATL = Automated Tape Laying.

AFP = Automated Fiber Placement.

HDLUP = Hand Lay-Up.

TBD = To Be Defined.

(1) The weight of dry fabric is 110 g/m².

(2) The weight of dry fabric is 310 g/m².

(3) Cure in oven.

(4) The material thickness shall be disregarded for cured part thickness determination.

(5) Considering one cured ply, manufactured by RTM (Resin Transfer Molding) process, with 59 ±5% fiber volume fraction.

(6) The thickness shall be considered just for process with 30 psi pressure.

(7) Cured in RTM (Resin Transfer Molding) process.

(8) Cured in VARTM (Vacuum Assisted Resin Transfer Molding) process.

(9) Weight range of 38 – 44 g/m², reference DUPONT™ TEDLAR® TWH10BS3 / POLYVINYL FLUORIDE FILM data sheet.

(10) TWH10BS3 is 1 mil thick, and TWH20BS3 is 2 mil thick. For reference, double the weight range for TWH10BS3 (see note 9).

(11) Resin Film Infusion (RFI) is an infusion process where carbon dry reinforcement stacks (MEP 15-063) are placed together with epoxy resin films (MEP 15-055) to obtain a polymeric composite ply, according to NE 40-237 process.

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5.3.11.3 Nondestructive Inspection

The acceptance criteria for internal discontinuities shall be according to Table 8 and the inspection method shall be according to NE 07-011.

The areas to be inspected and applicable quality classes (item 4.31) shall be indicated on the drawings through an inspection chart as exemplified in Figure 6.

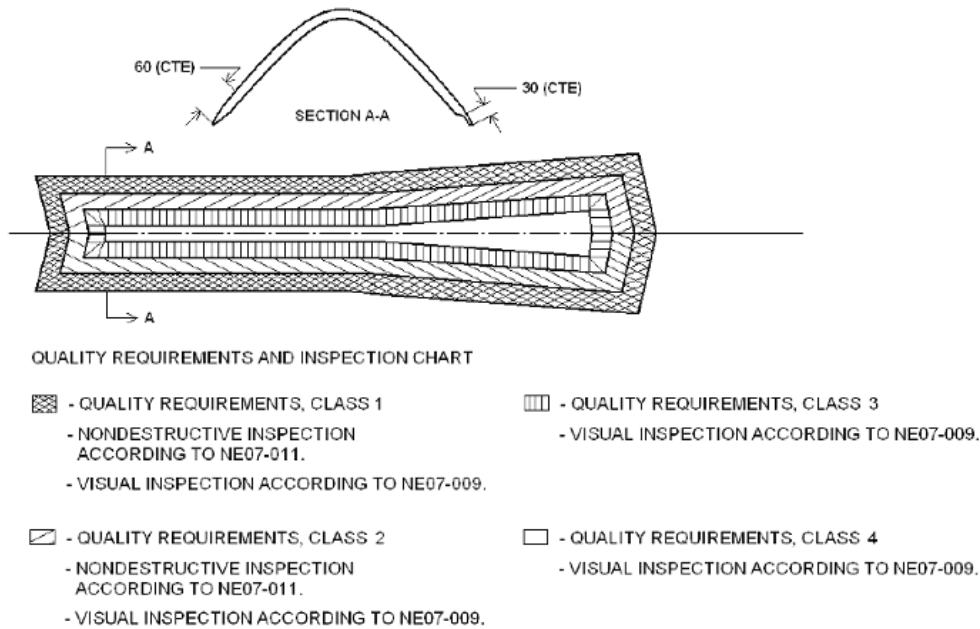


FIGURE 6 : NONDESTRUCTIVE INSPECTION CHART

When nondestructive inspection is not specified on drawing, the control of internal discrepancies shall be accomplished as follows:

- each part level I shall be inspected according to NE 07-011;
- translucent parts level II shall be visually inspected for internal discrepancies. Backlighting is acceptable to render the part sufficiently translucent for inspection. When necessary, nondestructive inspection according to NE 07-011 shall be used to investigate the nature of defects detected using visual inspection;
- when visual inspection according to item 5.3.11.3b is not possible, nondestructive inspection according to NE 07-011 shall be applied to level II parts at the discretion of Embraer's Quality and Materials Engineering's; and
- visual inspection according to item 5.3.11.3b or nondestructive inspection in accordance with NE 07-011 shall be applied to level III parts at the discretion of Embraer's Quality and Materials Engineering's.

5.3.11.4 Minimum Detectable Defects Size

The nondestructive inspection methods (other than radiography) shall demonstrate the sensitivity to detect and record at least the defects indicated in Table 7:

TABLE 7 : MINIMUM DETECTABLE DEFECTS SIZE

QUALITY CLASS	MINIMUM DETECTABLE DEFECT
1	6 mm DIAMETER CIRCLE OR 6 x 6 mm SQUARE
2	10 mm DIAMETER CIRCLE OR 10 x 10 mm SQUARE
3	13 mm DIAMETER CIRCLE OR 13 x 13 mm SQUARE
4	20 mm DIAMETER CIRCLE OR 20 x 20 mm SQUARE
SPECIAL (1)	HALF SIZE OF THE MAXIMUM ALLOWABLE DEFECT

(1) Applicable only when a special criterion is called out on drawing.

5.3.11.5 Acceptance Criteria for Manufactured Parts

5.3.11.5.1 The main types of defects for manufactured parts are given in Table 8, that shows the acceptable limits according the quality levels specified in the drawings and the recommended repairs.

5.3.11.5.2 The requirements established in Table 8 are independent of the adopted inspection form, which shall be specified on the engineering drawings.

5.3.11.6 Weight

Weight tolerance for manufactured parts is $\pm 5\%$ of calculated and released part weight.

5.3.11.7 Thickness

Nominal part thickness shall be determinate by the sum of each component individual nominal thickness at the considered area (see Table 6).

The acceptance criteria shall be according to tolerances defined on NE 03-047.

5.3.12 Manufactured Parts Storage

All parts, before the finishing primer application, shall be stored in an adequate area: free of dust, direct sun light ray's incidence, oil, grease, water or any other materials that may contaminate the part surface.

NOTE

The discrepancies defined in Table 8 shall not be only considered individually but also evaluated as "Multiple Discrepancies".

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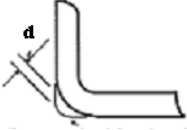
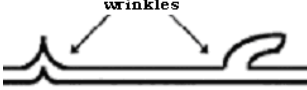
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TABLE 8 : ACCEPTANCE CRITERIA FOR FINISHED PARTS

DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
MULTIPLE DISCREPANCIES	MULTIPLE INSTANCES OF THE FOLLOWING DISCREPANCIES (IN ANY COMBINATION): MIS-ORIENTED FIBERS, CRUSHED CORE, INCLUSIONS, DELAMINATIONS, EDGE DELAMINATION, DISBONDS, VOIDS AND BRIDGING.	ALL DISCREPANCIES SHALL BE WITHIN THE ACCEPTABLE LIMITS FOR THE INDIVIDUAL DISCREPANCIES, AND THE MINIMUM DISTANCE BETWEEN DISCREPANCIES SHALL BE 10 TIMES THE LENGTH OF THE LARGER DISCREPANCY	THE REWORK LIMITS FOR THE MULTIPLE DISCREPANCIES IS THE SAME AS THE REWORK LIMITS FOR THOSE DISCREPANCIES AS INDIVIDUALS. THE DISTANCE BETWEEN UN-REWORKED DISCREPANCIES SHALL BE AT LEAST 5 TIMES THE LENGTH OF THE LARGER UN-REWORKED DISCREPANCY.	REWORK DISCREPANCIES AS INDIVIDUAL DISCREPANCIES UNTIL THE DISTANCE BETWEEN UN-REWORKED DISCREPANCIES ARE AT LEAST 5 TIMES THE LENGTH OF THE LARGER UN-REWORKED DISCREPANCY
SURFACE DISCREPANCIES				
EXTERNAL BRIDGE	– A REGION OF A CONTOURED PART WHICH HAS CURED WITHOUT PROPER COMPACTION AGAINST THE MOLD DEFORMING RADIUS AND CORNERS.  RADIUS REQUIRED BY DRAWING DIFFERENCE BETWEEN THE REQUIRED RADIUS AND ACTUAL CONDITION	MAXIMUM DISTANCE BETWEEN MOLD AND LAMINATE SURFACE (D) CLASS 1: $D \leq 0,5$ mm CLASS 2: $D \leq 0,7$ mm CLASS 3: $D \leq 1,0$ mm CLASS 4: $D \leq 2,0$ mm NO DISCREPANCIES ARE ALLOWED FOR SURFACES TO BE BONDED	ACCORDING TO ITEM 9.16 OF NE 20-038	
PLY WRINKLES	AN OUT-OF-PLANE DISTORTION OF FIBERS IN A CURED COMPOSITE PART, USUALLY ON THE BAG SIDE OF THE LAMINATE 	CLASS 1: NONE ALLOWED CLASS 2: NONE ALLOWED CLASS 3: WRINKLE HEIGHT $\leq 0,5$ mm; WRINKLE LENGTH ≤ 100 mm CLASS 4: WRINKLE HEIGHT $\leq 1,0$ mm; WRINKLE LENGTH ≤ 150 mm NONE ALLOWED IN FAYING SURFACES	MRB DISPOSITION	MRB DISPOSITION
SURFACE CRACKS IN RESIN	– ANY FRACTURE IN THE RESIN SURFACE	NONE ALLOWED	MRB DISPOSITION	MRB DISPOSITION
SURFACE SCRATCHES IN RESIN	– SCRATCHES AND SIMILAR DEFECTS IN THE RESIN SURFACE	ACCEPTABLE WHEN THE FIBERS ARE NOT DAMAGED AND/OR EXPOSED	ACCORDING TO ITEM 9.6.2 OF NE 20-038	

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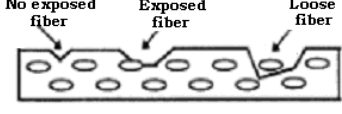
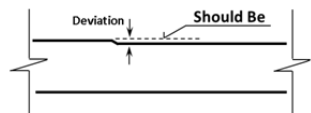
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SURFACE DISCREPANCIES				
RESIN SURFACE POROSITY/SURFACE RESIN STARVATION	- VOIDS OR PITS PRODUCED BY LACK OF RESIN OR BY ENTRAPPED AIR OR GAS 	ACCEPTABLE WHEN THE FIBERS ARE NOT DAMAGED AND/OR EXPOSED	WHEN THE EXPOSED FIBERS ARE NOT DAMAGED, LOOSE OR DRY	APPLY RESIN ACCORDING TO ITEMS 8.6 AND 9.6.1 OF NE 20-038
			WHEN THE EXPOSED FIBERS ARE DAMAGED, LOOSE OR DRY	MRB DISPOSITION
RICHNESS SURFACE IN RESIN	- EXCESS RESIN AREAS, USUALLY ON MOLD SIDE OF LAMINATES. CONTOUR OF PARTS IS NOT AFFECTED	MAXIMUM AFFECTED AREA OF PART PROVIDED PART WEIGH IS WITHIN THE SPECIFIED LIMIT CLASS 1: 20% CLASS 2: 30% CLASS 3: 50% CLASS 4: 80%	UNLIMITED	SAND WITH 150 GRIT OR FINER. DO NOT SAND INTO FIBERS.
SURFACE RIDGES IN RESIN/RESIN SURFACE LUMPS	- LOCALIZED RAISED ACCUMULATION OF EXCESS OF RESIN NOT CONTAINING FIBERS	- NONFAYING SURFACES: CUMULATIVE HEIGHT $\leq 0,5$ mm. - FAYING SURFACES: CUMULATIVE HEIGHT $\leq 0,1$ mm.	UNLIMITED	SAND WITH 150 GRIT OR FINER. DO NOT SAND INTO FIBERS.
SURFACE IMPRESSIONS IN RESIN	- IMPRESSIONS OR MARKS ON THE SURFACE PART EXAMPLES OF ACCEPTABLE IMPRESSIONS OR MARKS INCLUDE PEEL PLY SPLICE OVERLAPS, RELEASE FILM SPLICE OR AND MARKS OF TOOLING FINISHING.	ACCEPTABLE WHEN IMPRESSIONS OR MARKS $\leq 0,15$ mm DEEP WHICH ARE TRANSFERRED FROM TOOL SURFACE TO THE PART AND POSSIBLE TO BE ELIMINATED BY SANDING OR FINISHING PROCEDURES.	OVER THE TEFLON PLY: NONE ALLOWED	ACCORDING TO ITEM 9.17 OF NE 20-038
			ON OTHER MATERIALS: UNLIMITED	SAND WITH 150 GRIT OR FINER. DO NOT SAND INTO FIBERS.
DISCONTINUITY OF EXTERNAL LOFTING (STEP)	- SYSTEM LINE DEVIATION CAUSED BY TOOLING PARTITION 	CLASS 1: $\leq 0,15$ mm CLASS 2: $\leq 0,30$ mm CLASS 3: $\leq 0,50$ mm CLASS 4: $\leq 0,70$ mm	ACCORDING TO ITEM 9.20 OF NE 20-038. DEVIATION ABOVE THESE VALUES SHALL BE SUBMITTED TO MRB DISPOSITION.	

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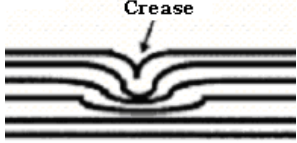
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DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
SURFACE DISCREPANCIES				
CREASES	– LINEAR DEPRESSION OF LAMINATE CAUSED BY PLY FOLDING. CREASES ARE USUALLY FILLED WITH RESIN 	CLASS 1: NONE ALLOWED CLASS 2: NONE ALLOWED CLASS 3: CREASE DEPTH $\leq 1,0$ mm OR 50% OF THE NOMINAL THICKNESS OF THE LAMINATE, WHICHEVER IS LESS. CLASS 4: CREASE DEPTH $\leq 2,0$ mm OR 50% OF THE NOMINAL THICKNESS OF THE LAMINATE, WHICHEVER IS LESS	ACCORDING TO ITEM 9.18 OF NE 20-038.	
EXPANDED COPPER FOIL WRINKLES	– A RAISED FOLD OF EXPANDED COPPER FOIL, NOT AFFECTING REINFORCEMENT PLYS (NO PLYS WRINKLES)	- NONFAYING SURFACES: ACCEPTABLE - FAYING SURFACES: NONE ALLOWABLE	UNLIMITED	SAND WITH 150 GRIT OR FINER. DO NOT SAND INTO FIBERS. REPAIR ACCORDING TO NE 80-060
ALUMINUM WIRE MESH WRINKLES	– A RAISED FOLD OF ALUMINUM WIRE MESH, NOT AFFECTING REINFORCEMENT PLYS (NO PLYS WRINKLES)			
EXPANDED COPPER FOIL TEAR OR ALUMINUM WIRE MESH TEAR	– ANY RUPTURE OF THE EXPANDED COPPER FOIL OR ALUMINUM WIRE MESH – ANY RUPTURE OF THE ALUMINUM WIRE MESH THREADS	UNACCEPTABLE RUPTURES: - RUPTURES LOCALIZED 25 mm OR LESS FROM METALLIC BOND STRAPS AND FIXATION POINTS; -RUPTURES WITH LENGTH LARGER THAN 10 mm. NOTES 1) EXCEPT FOR LACK OR BREAKAGE OF FOIL FROM THE EDGE WHICH ACCEPTABLE LIMIT SHALL BE ACCORDING TO THE ITEM BELOW "EXPANDED COPPER FOIL ABSENCE OR LACK". 2) FOR WING/FUSELAGE FAIRING, IT IS ACCEPTABLE A MAXIMUM CUMULATIVE RUPTURE OF 2500 mm² FOR EACH 1 m² OF AREA. 3) FOR WINDOWS AND ACCESS DOORS CONTOURS, RUPTURES OF THE EXPANDED COPPER/ALUMINUM WIRE OVER THE RECESS AREA IS ACCEPTABLE.	UNLIMITED	REPAIR ACCORDING TO NE 80-060

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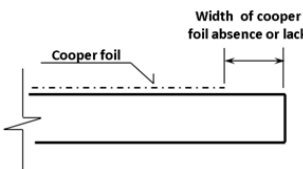
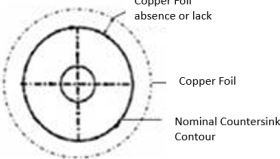
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DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
SURFACE DISCREPANCIES				
EXPANDED COPPER FOIL ABSENCE OR LACK FROM THE EDGE OF PART	– ANY ABSENCE OR LACK OF COPPER FOIL FROM THE EDGE OF PART 	ACCEPTABLE BREAKAGE OF UP TO 3 mm WIDTH ON THE COPPER FOIL FROM THE EDGE OF THE PART	BREAKAGE IN EXCESS OF 3 mm ON THE EDGE OF PART	REPAIR ACCORDING TO NE 80-060
LACK OF COPPER FOIL AROUND THE ATTACHMENT HOLES	– ANY LACK OF COPPER FOIL FROM THE COUNTERSINK NOMINAL DIAMETER 	HOLES AT EXTRA THICK COPPER FOIL AREA: NO LACK/ABSENCE ACCEPTABLE HOLES AT REGIONS WITH NO EXTRA THICK COPPER FOIL: A) GENERAL: UP TO 1 MM FROM NOMINAL COUNTERSINK B) PARTS WITH LESS THAN 10 HOLES: 1 HOLE WITH LACK UP TO 3 MM FROM NOMINAL COUNTERSINK DIAMATER C) PARTS WITH 10 OR MORE HOLES: UP TO 10% OF THE HOLES, WITH LACK UP TO 3 MM FROM NOMINAL COUNTERSINK LINE	UNLIMITED	REPAIR ACCORDING TO NE 80-060
EXPANDED COPPER FOIL SPLICE	– THE JOINING OF TWO SEGMENTS OF EXPANDED COPPER FOIL BY OVERLAPPING THE SEGMENTS OVERLAPPING OF TWO EXPANDED COPPER FOIL SEGMENTS JOINED	ACCEPTABLE, PROVIDED THE ELECTRICAL RESISTANCE BETWEEN THE SEGMENTS IS NOT GREATER THAN 25MILLIOHMS WHEN MEASURED ACCORDING TO NE 80-060.	UNLIMITED	REPAIR ACCORDING TO NE 80-060.
ALUMINUM WIRE MESH SPLICE	– THE JOINING OF TWO SEGMENTS OF ALUMINUM WIRE MESH BY OVERLAPPING THE SEGMENTS			
EXPANDED COPPER FOIL OR ALUMINUM WIRE MESH DISPLACEMENT	-	-	-	ACCORDING TO NE 80-060

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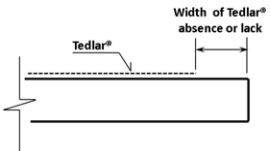
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DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
SURFACE DISCREPANCIES				
TEDLAR® WRINKLES/ BONDABLE TEFLON	- A RAISED FOLD OF TEDLAR® OR BONDABLE TEFLON	- NONFAYING SURFACES: ACCEPTABLE - FAYING SURFACES: WRINKLES HEIGHT ≤0,1 mm.	UNLIMITED	SAND WITH 150 GRIT OR FINER. DO NOT SAND INTO FIBERS. IF TEDLAR® OR TEFLON IS REMOVED DURING SANDING, REPAIR ACCORDING TO ITEM 9.10 OF NE 20-038
DAMAGED TEDLAR®	MISSING TEDLAR® AROUND CONTOUR OF THE PART FROM THE CONTOUR MACHINING PROCESS SCRATCHES, WHICH PROMPTLY REMOVED THE TEDLAR® TEDLAR® OR INCLUSION OF ANY MATERIAL NOT SPECIFICALLY CALLED OUT ON THE ENGINEERING DRAWING WHICH IS ADHERED TO THE PART SURFACE OR PLACED BETWEEN THE TEDLAR® AND THE LAST OR FIRST PLY OF THE PART. NO FIBER DAMAGE.	ACCEPTABLE WIDTH OF TEDLAR® ABSENCE OR LACK 5 mm AROUND CONTOUR EDGE OF PART  ACCEPTABLE AREA OF TEDLAR® ABSENCE OR LACK OF 50 x 50 mm, LIMITED TO THREE AREAS IN THE PRODUCT	UNLIMITED	REMOVE THE FOREIGN OBJECT AND REPAIR ACCORDING TO ITEM 9.10 OF NE 20-038. REMOVE CHARRED TEDLAR® BY SANDING WITH 150 GRIT OR FINER. DO NOT SAND INTO FIBER.
DAMAGED TEFLON	- TEFLON OR INCLUSION OF ANY MATERIAL NOT SPECIFICALLY CALLED OUT ON THE ENGINEERING DRAWING WHICH IS ADHERED TO THE PART SURFACE OR PLACED BETWEEN THE TEFLON PLY AND THE LAST OR FIRST PLY OF THE PART. NO FIBER DAMAGE	NONE ALLOWED	UNLIMITED	REPAIR ACCORDING TO ITEM 9.11 OF NE 20-038

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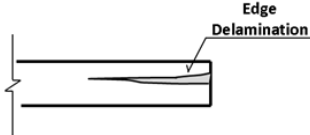
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SURFACE DISCREPANCIES				
SURFACE DEPRESSIONS	- DIMPLES, WATERMARKS, OR ORANGE PEEL EFFECT ON THE OUTSIDE OR BAG SIDE SURFACE	DEPRESSION DEPTH $\leq 0,1 \text{ mm}$ DEPRESSIONS ON NONFAYING SURFACES GREATER THAN OR EQUAL TO 0,25 mm DEEP ARE ACCEPTABLE PROVIDED THE LENGTH IS LESS THAN OR EQUAL TO 25 mm. BAG SIDE DEPRESSIONS RELATED TO HONEYCOMB CELLS NOT GREATER THAN DIAMETER OF CELLS ARE ACCEPTABLE	MRB DISPOSITION	MRB DISPOSITION
THROUGH PIN HOLE	- ABSENCE OF RESIN OR FIBER PASSING COMPLETELY THROUGH THE LAMINATE. VISIBLE TO THE NAKED EYE.	NONE ALLOWED	MRB DISPOSITION	MRB DISPOSITION
FIBER DAMAGE	- ANY RUPTURE OF THE FIBERS	NONE ALLOWED	ACCORDING TO ITEM 8.1 OR 9.1 OF NE 20-038	
DAMAGED EDGE	- LACK OF MATERIAL AT THE EDGE OF PART, FOR EXAMPLE DUE TO TOOLING ENTRANCE IN THE PARTS EDGE	NONE ALLOWED	ACCORDING TO ITEM 8.8 OF NE 20-038	
SHOR TRIM	- SHORTER DIMENSION	NONE ALLOWED	ACCORDING TO ITEM 8.2.1 OF NE 20-038	
EDGE DELAMINATION	- A SEPARATION OF PLIES FROM EACH OTHER THAT EXTENDS TO THE EDGE OF THE LAMINATE 	CLASS 1: $A + B \leq 20 \text{ mm}$ CLASS 2: $A + B \leq 25 \text{ mm}$ CLASS 3: $A + B \leq 40 \text{ mm}$ CLASS 4: $A + B \leq 60 \text{ mm}$ (2)	ACCORDING TO ITEM 9.3 OF NE 20-038 SINCE DELAMINATION AREA DOES NOT AFFECT THE ATTACHMENT OF THE PART.	
MIS-ORIENTED FIBERS	- IMPROPER PLACEMENT OF ADJACENT FIBERS IN A LOCALIZED AREA OF PLY 	CLASS 1: $A + B \leq 20 \text{ mm}$ CLASS 2: $A + B \leq 25 \text{ mm}$ CLASS 3: $A + B \leq 40 \text{ mm}$ CLASS 4: $A + B \leq 60 \text{ mm}$ (2)	MRB DISPOSITION	MRB DISPOSITION

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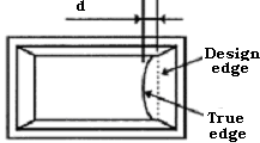
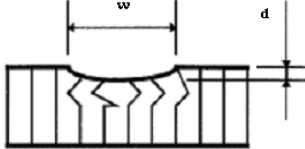
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TABLE 8 : ACCEPTANCE CRITERIA FOR FINISHED PARTS

DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
SURFACE DISCREPANCIES				
INCORRECT PLY LOCATION/ INCORRECT PLY SPLICE LOCATION	– PLY LOCATION OR PLY SPLICE LOCATION NOT IN ACCORDING TO DRAWING	NONE ALLOWED	MRB DISPOSITION	MRB DISPOSITION
CRUSH AT CORE EDGE	– CORE COLLAPSED LOCALLY IN WARD DESIGN EDGE RESULTING FROM SIDE PRESSURE DURING CURE 	MAXIMUM DISPLACEMENT FROM DESIGN EDGE OVER THE ENTIRE LENGTH OF PART EDGE CLASS 1: $D \leq 3$ mm CLASS 2: $D \leq 5$ mm CLASS 3: $D \leq 8$ mm CLASS 4: $D \leq 15$ mm	MRB DISPOSITION	MRB DISPOSITION
CRUSHED CORE	– LOCALIZED BUCKLING OF CELL WALLS. DENT IN SANDWICH LAMINATE SKIN. NO DISBOND OF LAMINATE SKIN FROM CORE 	CLASS 1: $D \leq 0,5$ mm $W \leq 25$ mm CLASS 2: $D \leq 1,0$ mm $W \leq 25$ mm CLASS 3: $D \leq 1,0$ mm $W \leq 50$ mm CLASS 4: $D \leq 2,0$ mm $W \leq 50$ mm	ACCORDING TO ITEM 8.10 OF NE 20-038	
BROKEN FIBER AT HOLES	– FIBER BREAKOUT AT PREPARED HOLES RESULTING FROM DRILLING OPERATIONS (DRILLING, COUNTERSINK, ETC.)	NONE ALLOWED	MRB DISPOSITION	MRB DISPOSITION
WARPAGE	DISTORTION OF THE PART BEYOND CONTOUR DEFINITION	44,5 N MAXIMUM FORCE TO MEET CONTOUR	MRB DISPOSITION	MRB DISPOSITION
SUPERFICIAL INCLUSIONS IN: -PART SURFACE -GLASS FIBER PLY OF GALVANIC PROTECTION -ALUMINUM AND COOPER WIRE MESH	– ANY MATERIAL ADHERED TO THE PART SURFACE OR BETWEEN THE PART SURFACE AND GLASS FIBER PLIES OR WIRE MESH NOT LISTED OR IMPROPERLY PLACED ACCORDING TO THE DRAWING	FOR ALL CLASSES: $A + B \leq 20$ mm (2)	FOR CLASSES 1 AND 2 AND DEFECT ON PART SURFACE: MRB DISPOSITION FOR CLASSES 3 AND 4 AND DEFECT OR GLASS FIBER PLY OF GALVANIC PROTECTION AND METALLIC WIRE MESH: $A + B > 20$ mm AND ≤ 40 mm	- REMOVE FOREIGN OBJECT - WHEN APPLICABLE, RESTORE LAY-UP ACCORDING TO NE 20-038 AND NE 80-060
			$A + B > 40$ mm	MRB DISPOSITION

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DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
INTERNAL DISCREPANCIES				
INTERNAL DEEP INCLUSIONS	- ANY MATERIAL ADHERED TO INTERNAL LAYER, NOT LISTED OR IMPROPERLY PLACED ACCORDING TO THE DRAWING, LIMITED TO A MINIMUM DEPTH OF 2 LAYERS FROM THE SURFACE	CLASS 1: $A + B \leq 20$ mm CLASS 2: $A + B \leq 25$ mm CLASS 3: $A + B \leq 40$ mm CLASS 4: $A + B \leq 60$ mm (2)	MRB DISPOSITION	MRB DISPOSITION
DELAMINATIONS	- A SEPARATION OF PLIES FROM EACH OTHER AND/OR FACING PLIES FROM CORE THAT OCCURS DURING CURE OR FINISHING OPERATIONS	CLASS 1: $A + B \leq 20$ mm CLASS 2: $A + B \leq 25$ mm CLASS 3: $A + B \leq 40$ mm CLASS 4: $A + B \leq 60$ mm (2)	MRB DISPOSITION	MRB DISPOSITION
DISBONDING / DELAMINATION / BREAKING / DISPLACEMENT OR LACK OF THE FIRST OR LAST PLY OF GLASS FIBER IN CARBON FIBER PARTS	BETWEEN GLASS FIBER AND CARBON PLIES: - DISBOND: ADHESIVE FAILURE BETWEEN THE PLIES - DELAMINATION: SEPARATION OF THE PLIES - BREAKING: GLASS FIBER PLY BREAKOUT - DISPLACEMENT: INCORRECT POSITIONING OF GLASS FIBER PLY - LACK OF GLASS FIBER PLY	NONE ALLOWED	ACCORDING TO ITEM 8.9 OR 9.9 OF NE 20-038	
DELAMINATIONS AT HOLES	- DELAMINATIONS AROUND PREPARED HOLES FOR FASTENER (EX: RIVET, PIN, SCREW, ETC.) INSTALLATION 	CLASS 1: ACCEPTABLE ONE AFFECTED HOLE WITH $D \leq 2,0$ mm IN ANY 15 ADJACENT HOLES CLASS 2: ACCEPTABLE ONE AFFECTED HOLE WITH $D \leq 2,0$ mm IN ANY 8 ADJACENT HOLES CLASS 3: ACCEPTABLE ONE AFFECTED HOLE WITH $D \leq 5,0$ mm IN ANY 5 ADJACENT HOLES CLASS 4: NOT MORE THAN ONE AFFECTED HOLE WITH $D \leq 5,0$ mm IN ANY 3 ADJACENT HOLES	MRB DISPOSITION	MRB DISPOSITION

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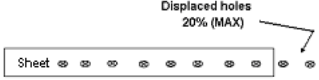
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DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
INTERNAL DISCREPANCIES				
ELECTRICAL BONDING STRIP HOLE DISPLACEMENT	– MAXIMUM DISPLACEMENT OF HOLES IN THE ELECTRICAL BONDING STRIP 	THE MAXIMUM ACCEPTABLE DISPLACEMENT IS 2,5 mm NONE ALLOWED WHERE THERE IS BONDING JUMPER INSTALLATION	UNLIMITED	REPAIR ACCORDING TO NE 80-060
DISBONDS	– ADHESIVE FAILURE RESULTING IN A DELAMINATION BETWEEN BONDED LAMINATES OR BETWEEN A LAMINATED AND CORE	CLASS 1: A + B ≤ 20 mm CLASS 2: A + B ≤ 25 mm CLASS 3: A + B ≤ 40 mm CLASS 4: A + B ≤ 60 mm (2)	MRB DISPOSITION	MRB DISPOSITION
VOIDS	– AIR POCKETS RESULTING FROM GAS ENTRAPMENT, LACK OF ADHESIVE, LACK OF ADHESIVE FOAM, LACK OF FILLER, LACK OF RESIN, ETC.	CLASS 1: A + B ≤ 20 mm CLASS 2: A + B ≤ 25 mm CLASS 3: A + B ≤ 40 mm CLASS 4: A + B ≤ 60 mm (2)	MRB DISPOSITION	MRB DISPOSITION
VOIDS ON POTTED DRIVED HOLES (HONEYCOMB)	– AIR POCKETS RESULTING FROM GAS ENTRAPMENT OR LACK OF RESIN POTTING FILLER	CLASS 1: NONE ACCEPTABLE CLASS 2: 10% OF DRIVED HOLE WALL SURFACE CLASS 3: 20% OF DRIVED HOLE WALL SURFACE CLASS 4: 30% OF DRIVED HOLE WALL SURFACE	MRB DISPOSITION	MRB DISPOSITION
LACK OF SINTETIC FOAM	SEALANT FAILURE OR ANY OTHER DISCREPANCY (VOIDS, DISBONDS, DELAMINATIONS, ETC.) EXTENDING FROM CORE AREA TO AN AREA EXPOSED TO THE ATMOSPHERE (CONSULT PROCESS 5 FROM NE 06-042)	-	MRB DISPOSITION	MRB DISPOSITION

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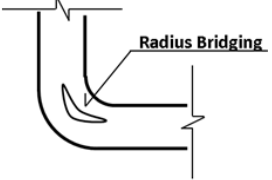
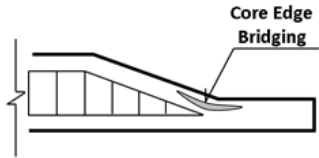
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DISCREPANCIES	DEFINITION (1)	ACCEPTANCE LIMITS	REPAIR LIMITS	REPAIR PROCEDURES
INTERNAL DISCREPANCIES				
RADIUS BRIDGING	– A CONDITION WHERE ONE OR MORE PLIES SPAN A RADIUS, STEP, OR CHAMFERED EDGE OF CORE CREATING A VOID BETWEEN PLIES 	CLASS 1: $A + B \leq 12 \text{ mm}$ CLASS 2: $A + B \leq 20 \text{ mm}$ CLASS 3: $A + B \leq 30 \text{ mm}$ CLASS 4: $A + B \leq 50 \text{ mm}$ (2)	MRB DISPOSITION	MRB DISPOSITION
CORE EDGE BRIDGING	– A CONDITION WHERE ONE OR MORE PLIES SPAN A RADIUS, STEP, OR CHAMFERED EDGE OF CORE CREATING A VOID BETWEEN PLIES 	CLASS 1: $A + B \leq 50 \text{ mm}$ CLASS 2: $A + B \leq 60 \text{ mm}$ CLASS 3: $A + B \leq 80 \text{ mm}$ CLASS 4: $A + B \leq 100 \text{ mm}$ (2)	MRB DISPOSITION	MRB DISPOSITION
LACK OF SEALING	SEALANT FAILURE OR ANY OTHER DISCREPANCY (VOIDS, DISBONDS, DELAMINATIONS, ETC.) EXTENDING FROM CORE AREA TO AN AREA EXPOSED TO THE ATMOSPHERE CONSULT PROCESS 6 FROM NE 06-042)	NONE ALLOWED	MRB DISPOSITION	MRB DISPOSITION
POROSITY	– VOID CONTENT OF THE LAMINATE BEYOND ACCEPTABLE LIMITS AS SPECIFIED IN THE APPLICABLE MATERIAL SPECIFICATION	INDIVIDUAL DISCREPANCIES: CLASS 1: $A + B \leq 50 \text{ mm}$ CLASS 2: $A + B \leq 60 \text{ mm}$ CLASS 3: $A + B \leq 80 \text{ mm}$ CLASS 4: $A + B \leq 100 \text{ mm}$ (2) TOTAL AREA OF MULTIPLE DISCREPANCIES IN ANY 100 cm² OF REFERENCED AREA: (3) CLASS 1: 10 cm² CLASS 2: 15 cm² CLASS 3: 25 cm² CLASS 4: 35 cm²	MRB DISPOSITION	MRB DISPOSITION

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MRB = Material Revision Board.

(1) Surface discrepancy: Any discrepancy visible to the naked eye.

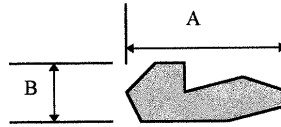
Internal discrepancy: Any discrepancy not in the part surface.

Internal discrepancies are not visible to the naked eye except when the part is translucent.

(2) A = length of the discrepancy.

B = width of the discrepancy measured perpendicularly to the length of discrepancy (A).

Any detected defect with A + B equal to or greater than twice the diameter or width of the applicable Minimum Detectable Defect shall be evaluated regardless of its width.



(3) The 100 cm reference area shall be a circle, a square or a rectangle containing the discrepancies. The location and shape of the 100 cm reference area shall be such that the worst condition is evaluated.

6. PROCESS QUALIFICATION

The process qualification is performed in order to guarantee that composite materials parts processor is able to replicate the properties obtained during the original materials qualification and design allowable determination and to show the ability to produce parts as designed in terms of properties and dimensions.

For so, process qualification is divided in three distinct phases: processor equivalency, First Part Qualification (FPQ) and First Article Inspection (FAI), which are detailed in next items.

NOTE

This qualification procedure is not applicable to wet lay-up production process parts.

6.1 PROCESSOR EQUIVALENCY

Processor equivalency is used to determine if a modified composite manufacturing process or new processor is able to fabricate composite panels with equivalent material properties compared to original qualified material.

The test matrix and procedures to perform the Processor Equivalency is defined by NE 40-267.

6.2 THERMAL PROFILE EVALUATION

For parts thermal profile, composite parts shall follow the cure cycle according to defined on part applicable standards and drawings. To demonstrate compliance and to define the control and monitoring thermocouple positions and guarantee that all curable materials successfully fits to the cure cycle, a thermal profile survey is necessary.

The part thermal profile procedure and acceptance criteria are defined in NE 40-251.

NOTE

Thermal profile of tooling shall be made during tooling qualification phase, before FPQ.

6.2.1 Tooling Thermal Profile Evaluation

Tooling thermal profile evaluation shall be performed before FPQ starts in order to indicate the leading and lagging thermocouple positions outside and inside the part lay-up area. The evaluation is not necessary if tooling fulfills all the following conditions:

- Superficial projected area less than 1 m²;
- Thickest area less than 7,5 mm for the composite tooling and less than 12,0 mm for the metallic tooling; and
- Engineering judgment evaluation based on tooling complexity.

In order to evaluate the tooling thermal profile, the thermocouples shall be located as follow:

- a) at least one thermocouple on autoclave atmosphere, oven atmosphere, plate; press or internal heated mold, as applicable;
- b) at least one thermocouple on the thickest tooling area, directly on the part lay-up region;
- c) at least one thermocouple on the thinnest tooling area, directly on the part lay-up region;
- d) at least one thermocouple on the thickest tooling area, out of the part lay-up region;
- e) at least one thermocouple on the thinnest tooling area, out of the part lay-up region; and
- f) at least one thermocouple representing the actual cure control thermocouple position for production parts. If the control position was not still chosen, various thermocouples may be used to define the best position (as close as possible to the lagging thermocouple).

After thermocouples positioning, for thermocouples within vacuum bagged region, apply an "Airweave S" breather or similar and a nylon vacuum bag, sealing the system and applying vacuum, pressure and temperature to simulate a whole cure cycle according to applicable specification.

NOTES

- 1) Only for tooling thermal profile cure load purpose, it is permitted to reduce the dwell times on the cure cycle to 15 minutes based on the lagging thermocouple.
- 2) For tooling thermal profile evaluation, the minimum heating rate on autoclave atmosphere thermocouple is 1,3°C/min.

6.3 FIRST PART QUALIFICATION (FPQ)

First Part Qualification (FPQ) procedure is performed on the first production part in order to verify that the production procedures, inspection techniques and the final part properties are in accordance with the design requirements, represented on drawings and engineering specification.

The FPQ process shall be performed again in case of any change in material, process and tooling, as described in NE 40-251.

NE 40-251 shows guidelines, procedures and requirements for FPQ process.

6.4 FIRST ARTICLE INSPECTION (FAI)

The First Article Inspection (FAI) shall be performed and approved in order to release the article for aircraft prototype and production of part and shall follow AS9102.

It shall be weighted, dimensionally inspected and measurements of thickness performed according to NE 40-251.

The production of FAI article may start before FPQ process is concluded since previously negotiated with Embraer and respecting the following conditions:

- Tooling and or part/tooling thermal profile successfully completed within the FPQ procedures according to item 6.2 and complying with success criteria described in NE 40-251.
- No problem that may cause FPQ rejection get detected during the lay-up, cure or any other required phases before FPQ thermal profile, such as nesting problems, tooling issues, etc.
- The process and resources (tools, lay-up and cure procedures) used to manufacture FPQ parts shall be the same for FAI parts manufacturing.

The final FAI and all parts fabricated (including represented parts) approval and delivery are conditioned to the FPQ final results report approval.

6.4.1 FAI Revalidation

The First Article Inspection (FAI) shall be performed and approved in order to release the article for aircraft prototype and production of part. FAI revalidation shall be according to AS9102.

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6.5 PERIODIC DESTRUCTIVE TESTS

Periodic destructive testing shall be performed in parts levels I and II when the nondestructive test (NDT) may not be performed only in areas which were not evaluated by NDT and that are considered critical (areas not evaluated by NDT) to the part structure, in accordance with Table 9.

The request for the periodic destructive tests shall be called in the drawing by NI 1392 - Perform the Periodic Destructive Test according to NE 20-032.

TABLE 9 : TEST SCHEDULE FOR PARTS

INTERVAL	PART
1	22 - 28
2	97 - 103
3	197 - 203
4	SUBSEQUENT, CONTINUE AT INTERVAL OF 100 $\pm$ 3

NOTE

Successful completion of one interval allows progression to the next higher interval. However, in addition to the appropriate corrective actions, failure at a particular interval, as a minimum, requires return to the previous shorter test interval.

6.5.1 Dissection in Periodic Destructive Tests

Dissection shall verify defects like porosities, delaminations or inclusions in areas where is not accessible to nondestructive testing, or that may not be adequately evaluated by same methods.

Those conditions arise typically from part features including but not limited to small radii, inaccessible co-cured spars and filler and sandwich structures.

7. QUALITY ASSURANCE PROVISIONS

7.1 MATERIAL QUALIFICATION

For normal using in production, the materials shall be qualified by Embraer Materials Engineering according to respective material specifications (see item 3).

7.2 MOLDS QUALIFICATION

7.2.1 Leakage Inspection

a) helium gas - the molds shall be inspected to check leakage, using helium gas before releasing to fabricate parts. No leakage is permitted; and

b) if helium gas is not available, the leakage check shall be performed using nylon film, breather and sealing system in order to apply vacuum of at least 81 kPa (0,81 bar), applying pressure and temperature according to respective part curing specification (NE) simulating a complete curing cycle. No leakage is permitted.

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8. GENERAL NOTES

8.1 RECOMMENDATIONS

8.1.1 Parts, after repaired, shall be resubmitted to Quality Control and Materials Engineering for approval, with the original quality level.

8.1.2 Marking of serialized parts shall be in accordance with:

- NE 01-107 for EMB-110/120/121;
- NE 01-114 for EMB-145;
- NE 01-121 for ERJ 170/190; and
- For no serialized parts shall be in accordance with NE 01-103.

8.1.3 Parts shall be inspected only after finishing operations.

8.1.4 Parts levels I and II shall have their documentation filed by Embraer Quality Control and Sub-Contractors for 7 years.

8.1.5 In regions where joints are not allowed, the drawing note NI 179 - Joint are not Allowed shall be used.

8.2 DRAWING CALLOUT

This Standard shall be called out on production drawings in the following form:

a) Embraer aircraft:

NI 176 - Overlapping Joint from <> to <> mm according to NE 20-032.

NI 178 - Butt Joint according to NE 20-032.

NI 200 - (**Deactivated for New Drawings**) Quality Requirements for Composite Material according to NE 20-032 Level < * >. See NI 1499.

NI 1392 - Perform the Periodic Destructive Test according to NE 20-032.

NI 1499 - Structural Criticality Level and Quality Class for Composite Parts according to NE 20-032, Level < * > and Class < ** >.

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Specify as follows:

* - LEVEL:

- Only one of levels I, II, or III defines the criticality level of part.

** - CLASS:

- Classes (1, 2, 3 or 4) for parts which request one quality class.

Ex.: <1>.

- Two quality classes for parts with areas which require different quality classes (in this case it is necessary define in the drawing the areas with different quality classes).

Ex.: <1 AND 2>.

NOTE

The drawings calling the NI 200 shall be interpreted considering the following equivalence:

- Parts Level I shall be considered Level I and Class 1,
- Parts Level II shall be considered Level II and Class 2, and
- Parts Level III shall be considered Level III and Class 3.

NI 1425 - Weight Tolerance according to NE 20-032.

NI 1426 - Thickness Tolerance according to NE 20-032.

b) AM-X:

Call flag note greater than 60. Example:

FL 61 - Requirements per NE 20-032 Level < * >, where:

* See item 8.2a.

8.3 BIBLIOGRAPHY REFERENCES

Abaris Training - Advanced Composite Structures: Fabrication and Damage Repair.

Aerospace Information Report (AIR) 4844.

ASTM D3878 – Standard Terminology for Composite Materials.

Composite and Metals Bonding Glossary.

Composite Material Handbook - Polymer Matrix Composites (CMH-17-1H).

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